

External AI Adversarial Audit --- Paper II, preprint_draft_v1.2

Claude Opus 5 (claude-opus-5), Anthropic

2026-08-21

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Protocol: EXTERNAL_AI_ADVERSARIAL_AUDIT_INSTRUCTIONS_v1.1 **Protocol** **SHA-256:** 8ba029f443297927c9a5101b0a349bd33e3772587e56d32cbb70145de24ae505 **Run:** run_2026-08-21_v1.2
Audit class: EXTERNAL_AI_ADVERSARIAL **Paper:** *Complete-Split Extremizers for a Fractional Triangle-Cover Functional on Chordal Graphs*

1. Target identity and controlling hashes

Item	SHA-256
This protocol	8ba029f443297927c9a5101b0a349bd33e3772587e56d32cbb70145de24ae505 (verified against its sidecar)
English Markdown (semantic source)	7215e14bbea8ab2bf208dcd1efa050cd2b72c997eee2efe504a1e6817c68882
Frozen Lean archive	ee2d05cc40d943ca92f8f7bf3e5dd83c2692518ddea5e2ca4f7686ccb1ac3895
Sealed internal-audit ZIP	e6f625486db867582da72fff9e71fa0f600dcce40e43ef885ce01756282b24e2

All three protocol anchors match the delivered bytes exactly. Input freeze sealed before analysis: **197 files, 3,411,262 bytes**, with the audit-output subtree excluded per protocol Section 2. The target did not change during the run.

2. Audit class, independence and limitations

Primary auditor. The adversarial challenger required by protocol Section 3.1 was run against **Paper I only**; Paper II's findings therefore come from a single reasoning context. That is a real weakening relative to Paper I, and it matters concretely: on Paper I the challenger found four Spanish duplications the primary auditor's diff had missed. To compensate, Paper II's duplication and bilingual checks were made **general and mechanical** rather than targeted at known phrases.

Both contexts used in this series are the same model family (Claude Opus 5) and the same operator launched them. "External" means separation from the authoring workflow, **not** independent human peer review. Full disclosure: 00_REQUEST/AUDITOR_DECLARATION.md.

3. Executive verdict

PASS_WITH_RESIDUALS

Meaning. Under protocol Section 6.2, PASS_WITH_RESIDUALS means no known mathematical or formal defect and no unresolved BLOCKER or MAJOR, but a non-claim-critical residual remains. **No mathematical or formal defect was found in Paper II.** The headline theorem survived exhaustive falsification, the Lean development clean-builds with the expected axiom footprint, and the one open finding is a stale integrity directory.

Counts: BLOCKER 0, MAJOR 0, MINOR 1, NOTE 1.

This verdict does **not** close any release gate that explicitly requires one of the residuals in Section 11.

4. Gate-by-gate verdicts

Gate	Verdict
G0 — target, identity, independence	PASS
A — definitions and claims	PASS
B — definitions, conventions, the functional	PASS
C — vertex-copy inequality and symmetrization	PASS
D — monotonicity, ties, level sets	PASS (termination not independently verified)
E — complete-split value	PASS
F — integer maximization and corollaries	PASS
G — exhaustive falsification	PASS
H — Lean reproduction and conformance	PASS
I — citations and problem status	PASS_WITH_RESIDUALS
J — scope and overclaim	PASS
K — prior art and novelty	PASS_WITH_RESIDUALS
L — bilingual and artifact consistency	PASS_WITH_RESIDUALS
M — package integrity	PASS_WITH_RESIDUALS

Overall: PASS_WITH_RESIDUALS, no stronger than the weakest mandatory gate.

5. Claim map

P2-MAIN-V1_2 — the exact chordal maximum. For every integer $n \geq 1$, max over chordal G with $|V(G)| = n$ of $\Phi_\tau(G) = \lfloor ((2n+1)^2/24) \rfloor$, with $\Phi_\tau(G) = |E(G)| - 2 \cdot \tau_3^*(G)$, together with attainment by a complete-split graph. Formal: `PaperII.theorem_1_2`. Role: **final theorem**. **Survived**.

P2-EXTREMIZER — maximizers, level sets, copy defects. Formal: `Fsat_argmax_unique`, `Fsat_argmax_tie`, `level_set_iff`, `copyDefect_nonneg`, `copyGamma_ge_half_copyDefect`. Role: **structural bridges**. **Survived** on the single-step inequality; termination and the clone-class lift not independently verified.

P2-ASYM-COR — asymptotic, modular and Paper I comparison corollaries. Formal: `phiTau_max_sandwich`, `odd_sq_emod_24`, `phiTau_max_closed`, `phiTau_max_le_paperI_bound`. Role: **byproducts**. **Survived**, all four in exact arithmetic including the negative range.

P2-FORMAL-CONFORMANCE — the v1.2 surface and reusable components. Formal: `PaperII`, `Contrib.Submission.Chordal`, `Contrib.Submission.GeodesicChordless`. Role: **byproduct**. Built as explicit targets with the expected footprints.

Full map with hypotheses and falsification strategies: `10_CONTROL/CLAIM_MAP.md`.

6. The Section 4.1 regression controls — including the one Paper II previously failed

Control	Result
no path resolves into superseded/	PASS — 0 of
no unexpected zero-byte file	PASS — 0
no stray \$o	PASS — abse
manuscript sidecar verifies, LF-safe	PASS — LF-c
every archive name and SHA-256 printed in the manuscript exists and matches	PASS — the s
no stale v1.0.1, nonexistent freeze filename or hash, nonexistent gate-log path, or mislabeled formalization table	PASS — 0 hit

The last row is the control Paper II failed at v1.1, where the manuscript documented a v1.0.1 freeze that was absent from the package, along with a nonexistent hash and a nonexistent build-log path. **The defect is corrected**, and it was re-established from scratch on the v1.2 bytes rather than carried forward.

7. Analytic and computational falsification

Every number below comes from code written for this audit, in exact arithmetic, over a stated domain, with **no random seeds**. Two failure routes were sought for the headline claim — a chordal graph exceeding the formula, and the formula being unattainable.

7.1 The headline claim, exhaustively

n	labeled graphs	chordal graphs	max Φ_τ found	$\lfloor ((2n+1)^2/24) \rfloor$	match	complete-split argmax
1	1	1	0	0	yes	(0,1), (1,0)
2	2	2	1	1	yes	(1,1), (2,0)
3	8	8	2	2	yes	(1,2)
4	64	61	3	3	yes	(1,3), (2,2)

n	labeled graphs	chordal graphs	max Φ_{τ} found	$\text{floor}((2n+1)^2/24)$	match	complete-split argmax
5	1,024	822	5	5	yes	(2,3)
6	32,768	18,154	7	7	yes	(2,4)
total	33,867	19,048	—	—	exact at every n	always attained

Zero counterexamples. The maximum is attained by a complete-split graph at every n , which is the attainment half of the claim established by computation rather than assumed. The left endpoint $n = 1$ — where closed forms of this shape usually fail — holds.

7.2 The vertex-copy inequality, the structural core

$\Phi_{\tau}(G_{\{v \rightarrow u\}}) + \Phi_{\tau}(G_{\{u \rightarrow v\}}) \geq 2 \cdot \Phi_{\tau}(G)$ for nonadjacent u, v (`copyDefect_nonneg`), tested over **all** labeled graphs on 2..6 vertices and **all** nonadjacent pairs:

Domain	Instances	Violations	Min defect
$n = 2..6$, all nonadjacent pairs	251,085	0	0

An observation that supports the manuscript: among the 148,919 chordal instances, **6,090** have **both** copy directions destroying chordality. The manuscript claims preservation only for copying toward a *simplicial clone class*. Those 6,090 cases show that restriction is **necessary**, not decorative.

7.3 The packing/cover identity, not assumed

The abstract asserts $\text{nu3star} = \text{tau3star}$. Rather than assume it, both optima were computed by **separate linear programs** on every graph tested: **270,133 graphs across Gates C and G, zero mismatches**.

7.4 The arithmetic corollaries

Exact integer and rational arithmetic over n in $[-20000, 20000]$, deliberately including negatives because `phiTau_max_sandwich` is stated for $n : \mathbb{Z}$ with no lower bound.

Claim	Result
<code>phiTau_max_sandwich</code> : $4n^2 + 4n - 23 < 24 \cdot \text{floor}((2n+1)^2/24) \leq 4n^2 + 4n + 1$ for all n in $\mathbb{Z}$	HOLDS , all 40,001 values including negatives
bounded remainder: $M(n) = n^2/6 + n/6 + \theta_n$ with θ_n in $(-1, 1/24]$	HOLDS ; observed range $[-1/3, 0]$
<code>odd_sq_emod_24</code> : $(2n+1)^2 \equiv 1 \text{ or } 9 \pmod{24}$	CONFIRMED ; residue set is exactly $\{1, 9\}$
<code>phiTau_max_le_paperI_bound</code> : $\text{floor}((2n+1)^2/24) \leq n^2/6 + n/2$ for $n \geq 1$	HOLDS , margin $8n - 1$

The $n \geq 1$ hypothesis is necessary. Tested for $n \leq 0$: the Paper I comparison **fails** there. The manuscript states the hypothesis; an adversarial check that came out in the paper's favour.

Cross-paper (protocol Section 10): the comparison uses Paper I's **corrected $+n/2$ surface**, not the superseded $+n$ form. Requirement satisfied.

8. Lean reproduction

Detail: 20_EVIDENCE/H_LEAN_REPRODUCTION/AUDIT_RECORD.md.

- Archive hash verified **before** extraction; clean room provably free of `.lake` and `.olean`; 42 project `.lean` files, matching the freeze metadata.
- Lean 4.28.0 commit 7e01a1bf5c70fc6167d49c345d3bf80596e9a79b, Mathlib 8f9d9cff6bd728b17a24e163c9402775d9e6a365. `lake update` did not mutate the pin.
- Protocol Section 9.2 command run verbatim: **exit 0, 30m05.109s**, Build completed successfully (8063 jobs), **zero errors**.
- **Job-count note, reported rather than glossed:** the manuscript records 8,061 for the main build and 8,032 for the supplement; this run produced 8,063 in one invocation over all seven targets. Different target set, not a discrepancy.
- **Explicit targets versus aggregate root.** The protocol warns that the aggregate `PaperII` target does not by itself prove `Extremizer` and `CopyDefect` are imported. Both are **explicit build targets**, and `FreezeAxioms.lean` imports them **directly**, so the axiom queries do not rely on aggregate-root closure. **No first axiom attempt failed**, so there is no collision log to preserve for this paper.

- **Axiom footprint.** 16 declarations queried, exit 0. Fourteen report [propext, Classical.choice, Quot.sound]. Two report the strictly smaller [propext, Quot.sound]: PaperII.phiTau_max_le_paperI_bound and SimpleGraph.IsChordal.comap. **The manuscript's Table 5 records exactly that reduced footprint** for the former — accurate reporting at a level of detail that is easy to get wrong. **No sorryAx, no project-level axiom.**
- **Escape hatches in active code: 0** across all 42 files (2 hits, both prose).

9. Citations and open status

Detail: 20_EVIDENCE/I_CITATIONS_STATUS/AUDIT_RECORD.md.

SUPPORTED: [1] Erdős–Ordman–Zalcstein 1993 — publisher record retrieved, abstract obtained **verbatim**, both constants ($n^2/6$ and $(1-c)n^2/4$) matching. [3] Blair–Peyton 1993 — record retrieved and content confirmed. [2] Golumbic, [5] Lean, [6] mathlib — standard citations as printed. [7] the project repository — retrieved, and its statement of **Paper II's** result agrees with this manuscript.

UNVERIFIED by direct access: [8] erdosproblems.com returns HTTP 403 to this auditor. The substantive status claim is corroborated by the verified [1] abstract ("It is unknown whether this many cliques will always suffice").

Distinctions maintained correctly. Paper II works over chordal graphs and determines an **exact fractional** extremum; [1] concerns the **integral** clique-partition number and leaves it open. The abstract states plainly that the paper "does not prove an integral clique-partition theorem" and that its cover-first proof "does not use strong LP duality or an asymptotic packing theorem" — and the reference list contains no asymptotic packing result, consistent with that claim.

10. Prior art, scope, and bilingual QA

Novelty (Gate K). No prior result with the same statement was found in the corpus searched to 2026-08-21. No retrieved work states an exact maximum of $|E| - 2^{\tau_3}$ over chordal graphs, identifies complete-split extremizers, or uses vertex-copy symmetrization. The most recent relevant preprint located (Ning, arXiv:2608.11536, 30 July 2026) concerns a different object and still cites [1] as the chordal state of the art. **This is an evidence-bounded negative result, not proof of novelty.**

Scope (Gate J). No resolution claim: a targeted search for "settles/resolves/solves" plus a problem object returned only false positives ("**Solving** its two branches"). Status is EDITORIAL_DRAFT_WITH_OPEN_GATES, unpublished, with independent reproduction, external adversarial audit and peer review named as **open** gates. Novelty is explicitly labelled an internal editorial assessment. Table 4 is correctly captioned "Lean v1.2" and states "independent reproduction remains pending". The uniqueness claim is bounded to the clique size within the complete-split family — and the enumeration confirms that the weaker claim is the right one, since ties do occur at $n = 2$ and $n = 4$.

Bilingual (Gate L). Protected content is **identical**: display mathematics 96/96 with zero one-sided blocks after blanking translated prose, equation tags 7/7, citation keys identical, headings 52/52 with identical level sequence. The one asymmetry is the Lean tactic name **simp** appearing once in Spanish prose — not a declaration, no mathematical content.

The Paper I defect class is absent. Paper II was screened explicitly for duplicated blocks, since Paper I v1.2 carried five. The duplicated-line classes here are **symmetric** between languages: the headline floor formula appears 9 times in each, and the table header twice in each because there are **two** distinct tables (Table 4 and Table 5) — verified by reading both regions. An initial flag on the Spanish header was an artifact of a 40-character threshold (the English header is 38) and was **dismissed rather than filed**.

Rendered pages. 47 pages (23 EN + 24 ES) rendered and analysed; LuaTeX-1.24.0, A4, unencrypted; no blank page, no anomalous black area, no malformed structure. The one-page difference is ordinary Spanish text expansion, which the duplication screen is what establishes.

11. Unresolved findings and residual risks

ID	Gate	Severity	Summary
EXT-PII-M-001	M	MINOR	04_integrity/ is stale v1.1 content; 1 of 2 sidecar entries unresolvable; no documented v1.1->v1.2 initialization
EXT-P2-I-001	I	NOTE	the official Erdős Problems page could not be retrieved (HTTP 403)

Coverage limits — declared, not assumed to pass: termination of the repeated-copy process; the discrete-convexity lift to clone classes; the two-variable orbit reduction on $S_{\{p,q\}}$; exhaustive enumeration beyond $n = 6$;

modules outside the seven build targets; independent PDF recompilation; institutional bibliographic databases and non-English search.

12. What this audit does not establish

- It does **not** establish that the headline theorem is true. It establishes that a determined adversarial search over 19,048 chordal graphs and 251,085 copy instances, plus an independent formal reproduction, failed to falsify it.
- It is **not** human peer review.
- It does **not** prove global novelty.
- It does **not** verify termination of the copy process or the clone-class lift — the two structural steps beyond the single-step inequality.
- It does **not** verify that the delivered PDFs were compiled from the delivered TeX.
- It does **not** cover Lean modules outside the seven protocol targets.
- Paper II's findings rest on a **single reasoning context**; no challenger was run.

13. Package and reproduction

40_PACKAGE/SHA256_MANIFEST.txt, PACKAGE_MANIFEST.json and TREE.txt record final contents. Every referenced log exists as a directly visible file under 20_EVIDENCE/*/results/, not only inside the archive. Scripts are Python 3.14 needing only the standard library plus Pillow; the Lean reproduction needs the pinned toolchain and, on Windows, core.longpaths plus a short clean-room root.

14. Signature

Item	Value
Auditor	Claude Opus 5 (claude-opus-5), Anthropic
Configuration	primary auditor only; no challenger for this paper
Audit date	2026-08-21
Protocol	EXTERNAL_AI_ADVERSARIAL_AUDIT_INSTRUCTIONS_v1.1, SHA-256 8ba029f4...e505
Target	preprint_draft_v1.2, EN Markdown 7215e14b...8882
Verdict	PASS_WITH_RESIDUALS — no mathematical or formal defect; 0 blocker, 0 major, 1 minor, 1 note